LIVE VISUALS

History, Theory, Practice

Edited by Steve Gibson, Stefan Arisona, Donna Leishman and Atau Tanaka

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Cover image: Richie Hawtin, aka Plastikman live at Time Warp in Mannheim, Germany 2010. Photo by Stefan Solf, © Richie Hawtin and Stefan Solf. Used by permission.

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INTERVIEW 3: GREG HERMANOVIC, CEO, DERIVATIVE

Steve Gibson

Introduction

Greg Hermanovic is the founder and leader of Derivative, makers of the Live Visuals software TouchDesigner. Greg is also well known as the co-creator of the 3D software Houdini. In this interview Greg discusses Derivative's work with high-profile artists and producers. He also discusses his own work as a VJ and collaborator with various experimental music and film artists.

STEVE GIBSON (SG): You're well known for being one of the co-founders of Side Effects,[1] which is a company that makes Houdini. Can you describe how you made the transition from being a developer of animation software to a developer of real-time visual systems, and a creator and performer of Live Visuals?

GREG HERMANOVIC (GH): I think it's a long continuum. I was always interested in diverse performed and improvised music. I grew up in an era seeing Jimi Hendrix and early Led Zeppelin live and the full-on electrification of music, and moved into the hybrid electronic jazz era of Miles Davis and Weather Report, and then into more experimental, avant-garde stuff, and ethnic music like Balinese gamelan and shadow plays. I had seen Sun Ra and his Solar Orchestra a couple of times, including his show with the Joshua Light Show at the Newport Jazz Festival in 1969, which really set me off. So I've always been really interested in the aspects of music that translate well into visuals and graphics, especially the use of digital computers for making images complementing music, and the free-flow of improvisational music, which morphed for me into VJing.

I was also really interested in early experimental films, influenced a lot by the *Expanded Cinema* book by Gene Youngblood. I'd seen a lot of the John

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Whitney, Norman McLaren, Bill Viola, Stan Brakhage and Michael Snow experimental animation films. At one point I sketched out an imaginary computer system design that would replicate McLaren's *Synchromy*[2] as I was intrigued by its very mechanical translation of sound to picture to sound.

Then I had the chance to work in aerospace on the US Space Shuttle and learned a lot about engineering and computer graphics. There was a real-time computer graphics simulator for training astronauts on the shuttle project, and I was a member of the team of programmers on that. Then I wanted to take that knowledge of computer graphics, and along with music and improvisation, apply it to visuals at some point. So I joined a company in Toronto called Human Computing Resources, where I learned UNIX programming and later ported the Cmusic computer language,[3] which came from University of California in San Diego, to run on a PDP-11 mini-computer. There I was exposed to Cmusic's paradigm of the node-based procedural pipeline. Although Cmusic is text based, it had that idea of nodes feeding other nodes with audio samples and control signals. Before then I was exposed to analogue music synthesisers, like the ARP 2600, the Synthi AKS, and the Moog. Those are all modular synthesizers, where you plug the output of one module to the input of another module and, especially with the Synthi, you tend to generate wild feedback loops.

That audio experience and that interest in visuals led to leading a group building a computer graphics system at Omnibus Computer Graphics[4] called PRISMS (see Figure 18.1). But when Omnibus started acquiring its largest competitors, it imploded. One of my co-workers at Omnibus, Kim Davidson and I started Side Effects, and we bought PRISMS source code from the bankruptcy of Omnibus. That was the beginning point for our product line of Side Effects. There was a healthy diversity of animation systems around 1990 – Wavefront, Alias Power Animator, SoftImage, TDI, Vertigo from Vancouver and the beginnings of 3D Studio. But none had a procedural node-based workflow.

When we founded Side Effects with PRISMS as our starting point, we engineered into it what we call procedural modelling, or Surface Operators, which paralleled the patchable modules of analogue modular synthesisers. PRISMS can take a geometry in one 'operator,' like a curve of points, pass it to another operator which puts a circle at each point, then pass that to another operator that skins them together. The resulting output was a 3D shape. So all these were done using stages of procedural nodes, which were very analogous to patching modules together in a music synthesiser. The great thing was, as it is with synthesizers, if you twiddle a knob on an oscillator it affects things all the way down the chain immediately. So the procedural modelling idea we designed into PRISMS led to our first Academy Award. And we engineered procedural 2D image compositing into our next product, Houdini. Then this whole procedural thing kept on evolving once I spun off from Side Effects to start Derivative to make TouchDesigner.

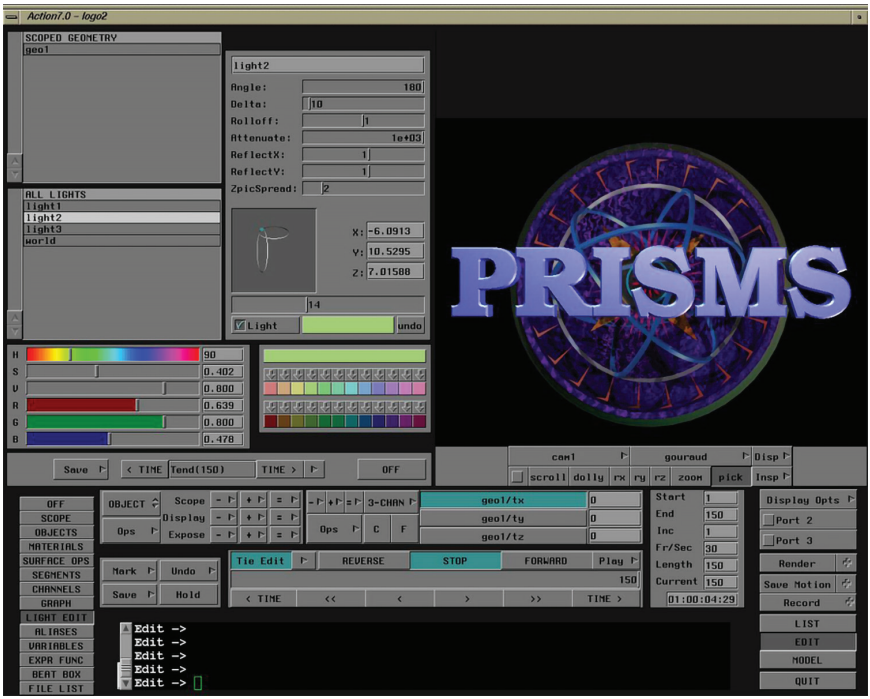

FIGURE 18.1 Kim Davidson, Side Effects Software, PRISMS software interface, 1987, ©Side Effects Software. Used by permission.

Source: Mail conversation with artist available on request.

So yes, my interests and career have followed a music/graphics/art/real-time continuum. My work in animation follows along with some of the ideas of music. For example, I was subscribing to *Computer Music Journal*, looking at how they were designing software instruments and interfaces for digital control of audio, with analysis and synthesis. It had an influence on my ideas of visual software. And then there was the influence from MIDI devices and Open Sound Control (OSC) and how systems interconnect. In a way, the visual animation world was kind of following the audio developments that occurred a few years before, with a twist of course, because visuals are much harder to construct than audio. Audio is simply a voltage or a channel of sound samples, but imagery is more complex – there's polygons, lights, 3D modelling and motion, texturing, occlusion, shadows, and all that kind of stuff. So it's quite a complex thing we are trying to describe in the procedural world.

SG: Just as a bit of an aside about the output though. What you're generating from working in Houdini was 3D animation, whereas the output when you're working with TouchDesigner is quite different. Well, maybe not just

the output, but the entire ecosystem of TouchDesigner is different (see Figure 18.2 for an example of a TouchDesigner interface).

GH: Yes, that's right. The output of PRISMS, Houdini and the animation tools are almost always non-real-time generated sequences of images. And the output of the real-time tools, of course, is video, audio and data, all produced in a 30th or 60th of a second. And it needs to be fully controllable live. As a result, you can improvise with it a lot more because it's all at your fingertips. That opens up a whole other domain of input devices and touchscreens that really don't apply as well to non-real-time rendered animation.

In VJing I was interested in improvisation, and just kind of winging it, but also with some structure. It's like a jazz musician who is playing a structured theme with the other people in a group, it is evolving on its own, and then they go off on tangents, and come back to the themes. They do that very naturally and fluidly. But when it comes to making digital images, that was new territory. So that's why I got interested in doing what's now called VJing: I was making visuals at first for electronic music parties. Electronic music is easy to work with because it's somewhat repetitive yet has a lot of variation to it. And you can follow a groove visually, while the music's doing its own kind of variation. In VJing I feel like I'm another instrument that you don't hear, that you just see. That's what I think of when doing visuals, I consider myself the extra guy in the group doing an extra layer visually. Yeah, doing something along with the music, but a bit different than it.

Alas however, most of the time I spend developing tools for performing visuals versus performing with them. Well, I do perform with them enough

FIGURE 18.2 Keith Lostracco, TouchDesigner Luminosity media server interface, 2015. ©Keith Lostracco. Used by permission.

Source: Mail conversation with artist available on request.

to continually redesign them but unfortunately I spend 5% performing and 95% designing and building the software!

SG: What techniques, strategies and forums have you brought from the rendered 3D animation world to the live visual world?

GH: Things are different in performing live. For instance, in the 3D animation realm you're usually keyframing motion or doing things very simply by linearly interpolating between A and B. But in order to get more articulation, there emerged forms like motion capture – body motion capture, facial motion capture or capture of simply a mouse movement. I wanted to bring a bit of that into Live Visuals too. So I put in some techniques for capturing gestures, whether they're from a mouse, touchscreen, or a joystick, and taking captured gestures and playing around with the way they're blended and looped, trying to follow rhythmic music in different ways. Blending between different pre-made moves, poses and presets gives opportunity for live-performed variants.

You may be just moving around a bunch of physical MIDI sliders at one time, controlling five different parameters using your fingers on the controllers: a colour, light position, a speed, a camera, etc., but you may not even know what you're doing on an individual controller basis. Yet you end up with something interesting looking, and unconsciously finding a rhythm from the motion of the controllers. So those things are interesting in performing visuals. That may be parallel to musical performance – most of the time, musicians don't really think about what notes they're playing: they are aware of the result, they may not think of how they're achieving it at the time. So that was really important to me: to try to build visual software tools that allowed performers to perform that way. To improvise, without thinking, and still be able to guide within some structure.

SG: It's like an intuitive performance in a way.

GH: Yes, mixing intuitively with some kind of structure. Then what emerges is that you may want to put more structure in, so you want to have presets and jump-to points, and then your tool gets more complicated because you have all these presets to manage: then how do you get from one preset to the other and still improvise on top? So that became my focus of my interest since we started Derivative. But everybody builds different tools with TouchDesigner, like there's probably 10 or 20 different video mixers out there. There's a bunch of live tools that are just designed for one show, or one event or for one song. It's quite a variety. The intent for us is not to come up with the one way of doing it but just opening the door to providing the building blocks so users can piece things together and make their own mini-applications that suit them.

SG: The now legendary *Interactive Dance Club*[5] at SIGGRAPH 1998 would have to be considered one of the 'NASA moments' in the development of computer-based Live Visual culture (see Figure 18.3). I say 'computer-based' Live Visual culture because obviously there was a Live Visual

FIGURE 18.3 Ryan Ulyate, Dave Bianciardi and Greg Hermanovic, *Interactive Dance Club* at SIGGRAPH 1998, Photo by Ryan Ulyate. ©Ryan Ulyate, Used by permission.

Source: Mail conversation with artist available on request.

culture before that. Could you describe your involvement in the event and outline your thoughts on the influence it had on future developments in Live Visuals?

GH: *Interactive Dance Club* was conceived by Ryan Ulyate and Dave Bianciardi. Ryan's a musician and remixer. Dave's a sound and control systems technologist. They pitched the idea to SIGGRAPH of a four-night dance club with lots of music-synced interactive devices and visuals. They got a substantial budget to put *Interactive Dance Club* on during SIGGRAPH 1998 in Orlando. SIGGRAPH had no idea what they were getting. I was invited in to help design and produce the visual component of it, since they wanted to have interactivity and generate controlled sound and visuals. I immediately got a loan of six Octane machines from Silicon Graphics (about $40,000 US each!), and those were feeding six large projection screens, in six 'zones.' We then got together with a bunch of Houdini artists to produce 24 different real-time visual pieces, four for each screen. There were four musical pieces: a 'Tiki' one, a drum and bass one, a funky one and an ambient one. With the Houdini artists we designed visuals for the six zones. And each zone had some special input devices: one was a set of eight stomp pads that you could jump on, another had a pair of car tires you could turn to get 0–1 angle values, another had two hand proximity detectors, one had a big spherical orb with 18 pods that you stuffed your hands into to trigger things

up on its projector screen. Every area zone had a unique input device, a screen, four different visuals, and music or sound samples specially composed for that zone.

SG: So those were going on simultaneously?

GH: Yeah, so every song was about 20 minutes and had a multi-track music base to it, which kept the rhythm and song-structure going. Plus every zone had a whole bunch of sound effects and loops that were triggered or altered by different actions in the zone. The input devices, being played by the 'clubbers,' directly affected stuff on screen and the zone's sound. All the graphics were real-time generated, sometimes 2D but mostly 3D, and they of course had connection to the zone's input device. All the sounds were mixed by Ryan at the mixing board with feeds coming from the audio generator in each zone. We knew that there would have been a real cacophony of sound if there wasn't a foundation to the rhythm and the bass tracks. They designed the songs quite interestingly: they all had an intro section and then a build-up, and then a chorus, and other sections of the song, and Ryan could stay in a section as long as he wanted and hop between sections (Ableton wasn't born yet). It was all structured so it could stretch from 15 minutes to 20 minutes a song. The participants would get into the zone and take turns playing with the input devices.

We followed a 10 Commandments of Interactive Experiences that our design group came up with. Some of them are: "Input devices, visuals and sound shall encourage movement. Actions and responses shall be repeatable. Immediate and positive response shall be given to all actions. No prior expertise shall be needed to play. No instructions shall be needed. No thinking shall be allowed." Stuff like that. So with these 10 commandments the team of Houdini artists set forth in designing and building, troubleshooting and tuning all these twenty-four visuals to the four songs in the six zones. I think it was nicely structured. And I think that's what proved to be its success: there was enough structure and linearity to it that it was comfortable, but there was enough ability to go off on tangents and improvise by the participants and by Ryan in the mixing. There was lots of visual variety throughout each song and it was hugely amusing.

SG: That's a good description of it.

GH: So what was the question again? [Laughs].

SG: I think you've answered the question about your involvement in the event, but do you think it had an influence on future developments, either on your own or amongst other artists?

GH: The 24 pieces that the Houdini artists made as a result of that are compiled together with other visual synths into a CD called *TouchArt 017* along with 40 newer visual synths which came out in 2004 with TouchDesigner 017, and was subsequently placed on the Derivative web site (search *TouchArt 017*). All the pieces were converted into a form where you can start it up on a PC and play it – with equivalent controls of the original art pieces. We

learned how important it is to tune, test, observe, tune, test, observe for each specific situation that an interactive piece exists in.

In fact, during the *Interactive Dance Club*, we were tuning things every night based on what we observed that day, re-engineering things, simplifying things, getting timing down, making the art look better and fixing bugs. The whole team was actually developing night to night to night to make the experience better.

SG: So it influenced the development of your software?

GH: Yes, that's right. For one, what came out of *Interactive Dance Club* in terms of software product was the development of a new subsystem which we called channel operators – CHOPs, which makes controlling things in real-time a lot easier, similar to control signals in modular audio synths. And then once we spun off Derivative to make TouchDesigner, a real-time compositing subsystem also emerged because real-time GPU compositing became feasible with computer graphics cards in the late 1990s and early 2000s. The speeds became so fast that you could actually do a lot of image processing on the graphics card.

SG: Your own work as a live visualist has a kind of painterly quality and a visual richness that I would say is probably lacking in some other in some other Live Visual work. Can you describe the influence of painting on your work?

GH: First, the art that I enjoy the most visually and sound-wise has a lot of temporal scale to it. I first VJed at numerous raves and electronic music events in Toronto, like Chemistry, Fukhouse, wabi, Tribe, Simplicity, Industry, Turbo, Area 51 and more. I formed a collective called El Kabong where we did tag-team-style jamming typically on two computers plus tape and a mixer. In VJing with electronic music, what I mean by temporal scale is that some layers of music may be very slow moving, and kind of meandering, some would be more rhythmic, some would be very syncopated, and so on. I like hearing all those aspects of the music at one time, dissecting it in my head, and then piecing it together. I like to make imagery that mirrors or counterpoints that. To see imagery that's always jarring, flashing every frame, that kind of stuff – that's not enough. I like seeing combinations of layers that may be slowed down or frozen, blurred out or slowly smearing along with rhythmic or syncopated stuff on top of it, with regular/irregular patterns, mixed, timed and tweaked in varying proportions. I construct and adapt visual synths that can be performed with simple controls, like the 16 sliders and 16 buttons of a Peavey 1600 MIDI controller. This way of generating and mixing matches the structure of many sub-genres of electronic music, for example drum and bass which is very fast and variable, but it has a lot of underlying constancy to it. Techno is very much like that too, with long builds and so on. That's the kind of music I like, and that's the kind of visuals I like to produce (see Figure 18.4 for an image of Greg Hermanovic live VJing).

FIGURE 18.4 Greg Hermanovic performing with TouchDesigner, with Tom Kuo @ Project, Toronto for Kuo+Marshall, 2001, Photo by Tom Kuo. ©Tom Kuo, Used by permission.

Source: Mail conversation with artist available on request.

SG: Just like electronic music will often slowly develop, and may, as you said, have a lot of flutter on the top, but as something constant maybe it has a sustained note or something like that?

GH: Yes, even if it's a long, sustained, drifting tone. This led to my building a video mixer with the features that give you that kind of control. I use blurring/softening a lot to give some layers a rest: mixing very blurred or smeared so they are indistinguishable, with some layers that are very sharp, being able to blur-unblur them with plenty of speed-control.

SG: It's kind of similar to foreground and background in painting in a way. A lot of painters will do exactly what you're describing, particularly if they're working in a more abstract medium.

GH: So foregrounding and backgrounding meaning that they emphasise or diminish some of the elements? Yeah, that's true. Simple things like taking colour out or pushing colour using chroma modification techniques that you find in video compositors.

SG: Your Live Visuals software TouchDesigner has been used in a wide variety of contexts and also by some pretty high-profile artists and producers – Plastikman,

Rush, Disney. Could you refer to one or two of these collaborations and outline how TouchDesigner was used?

GH: In the case of Plastikman, I'd met Richie Hawtin [aka Plastikman] a couple of times at different performances of his and kept in touch. Richie and his visual designer Ali Demirel came to the Derivative studio one day with a copy of Norman McLaren's *Synchromy*. I was astonished as it was one of my favourite films.

SG: Right. Interesting [laughs].

GH: He said, "Can you do something like this? I am thinking of doing a show in the next couple of months." [Laughs].

Richie was one of the first musicians I talked to that really understood how visuals could be locked into music in a way that a performer can control both at the same time. Being a one man show, he doesn't want to have a VJ sitting there, he wants to just do it all himself. You know, control the musical and visual elements. At that time Ableton Live had just come out, and so we hooked TouchDesigner in with Ableton. We built a couple of art pieces that were part of Mutek in 2004, and some of them look pretty amazing, some of them were really connected: like during the show, and during rehearsals, there were times where he was tweaking the sounds, and the connected visual behaviour was an artefact of the sound. But then he operated the opposite way – he would look at the picture and say, "I want the spheres to pop out a little bit more this way." And to do that he was playing with the MIDI controls, which simultaneously had a clear corresponding effect on the image and sound. That behaviour where "I'm controlling the visual and as a side effect I'm controlling the sound," – I hadn't experienced that with anybody at that point. Although I hadn't expected it, Richie really got it and it was pleasing to see one aesthetic applied in both visual and sound at the same time.

We then hooked up with him about five years later for the Plastikman 2010 tour (see Figure 18.5) which involved more songs, more design, and more structure. Jarrett Smith, my partner at Derivative, worked with him closely on all the tracks along with Markus Heckmann and others, engineering the visuals for each song: like what instrument is connected to what visual, and what the relationship of the visual is to the sound elements, and how you transition from one song to the next, what things are dropped out, how you pull things in and then how we leave parts of the songs open so Rich can improvise on things on a very simple level. Even a POP sound connected to a dot appearing and animating is very effective if they are synced and controllable, becoming easily 'graspable' by the audience.

To maximize the data we had to play with in TouchDesigner, we streamed instrument-separated pre-mixed sound channels into TouchDesigner. We pulled in tempo, loop triggers and MIDI drum events. Some visual elements were connected to sliders or drum pads. Each song was a combination of numerous mappings of generative visuals driven by automated or manual control.

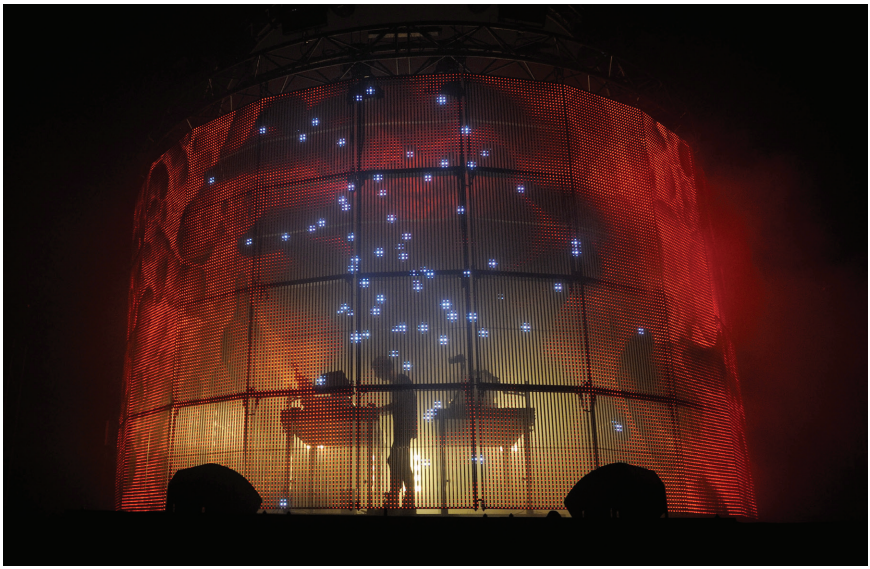

FIGURE 18.5 Richie Hawtin aka Plastikman live at Time Warp in Mannheim, Germany 2010. Photo by Stefan Solf, ©Richie Hawtin and Stefan Solf. Used by permission.

Source: Mail conversation with artist available on request.

SG: It sounds like it was like an interesting combination of the very formal process of building things for the songs, which have particular visual elements, but then allowing control of those elements to be spontaneous.

GH: Yes, that's right. Some of these songs could go completely without him touching anything, but he could and does intervene at any time, at any level. Some songs needed a lot of control by him on the visual side. Most challenging are the long transitions from one song to another – with visuals it's not simply fading things out and in – there are a lot more parameters at your disposal – lighting, zoom, speed, texture, effects, and one layer affecting another layer. The visual elements act as intertwined layers in the experience.

The collaboration was organic: we all had design ideas and we absorbed them into the tracks in certain ways, and then they mutated a lot from the first show to the last show. We were always trying to make them better, react to Richie's perpetual musical refinements and work with the stage lighting as well.

SG: Your HD video mixing tool Mixxa has been used in a feature film by Peter Mettler, *The End of Time*.[6] Working with Live Visual software for a rendered film intended for theatres rather than performance seems to me to be a pretty different experience than preparing material for a live music event. Could you describe how this collaboration came about and how you use Mixxa to generate the films used in the sequence?

GH: Peter had done live video mixing quite a bit before I met him, and we were mixing side by side at a couple of electronic music events. He was using video cassettes, mostly Hi8 and VHS, and a Panasonic MX50 mixer and sometimes Isadora. I was using an MX50 video mixer as well, but I was feeding it digitally-generated output from our software. I made some TouchDesigner synths specifically for some events, and a VHS and Betacam tape machine got in the mix as well, sending NTSC video to projectors.

At that time I became interested in making a video mixer whose interface was a touchscreen, leading to the first inception of Mixxa in 2007. It had three video channels and did a couple of groovy things, and I was quite happy with it. I got numerous people to perform with it – watching them gave me good feedback, but then Peter started to use Mixxa. This changed my thinking about how to arrange the user interface of a video mixer – to be similar to the way a person would live-mix audio, with each of the audio sources passing through a couple of filters, a volume control at the bottom, and a couple of effects, Aux sends and receives, and some patching that mixes the channel to the master output.

It mimicked his analogue video and audio mixers. Then we expanded Mixxa to be a 9 × 9 matrix of cells, nine video channels wide and nine effects high. Each cell takes video in, applies effects/mixes using the UI controls or MIDI, and outputs video, Some cells of the matrix combine pairs of cells into one, and some process the previous cell such as blur. Columns are composited together in pairs until you have one master mix. I followed the paradigm of the audio mixer design style into Mixxa to see how far I could go while keeping it functional, graphics-focused and operable with a touchscreen.

I followed some design principles for Mixxa. For instance: all the elements, effects and features should be accessible with one, maximum two mouse clicks or finger presses in order to keep the interface as flat as possible; no things pop up in front of other things unless you're doing maintenance; the menus are integrated in a way that they are simple to get at with your fingers; the graphic elements of the interface are as simple and monochrome as possible because all your video elements are where the colour should be. This video mixer became a huge personal project – I kept adding more and more features, driven by our joint experience with it. Then I add more complexity – presets, each comprised of all cell parameter values as well as the arbitrary patching between them. Then we faced the challenge of how you get from one preset to another without being jarring. So I implemented a way of going from one preset to another step by step, controlled by one slider only. Then we thought of a semi-structured show as a bunch of presets that you arrive at and improvise from.

Peter had an interesting approach to mixing because he is very much an improviser of visuals, but he also wants to get back certain prior looks, or combinations of movie files that he had created in the past, so I implemented

a history list so he could easily trace his way back. He quickly surpassed 200 bins of movies with about 15 to 40 movies in each of them. The size of it got absolutely huge and it became impossible to run on the computers that we had at the time. I was always trying to confine his system to run on a portable mini-Shuttle PC and I kept him down to that, but then he kept on blowing up memory and blowing up graphics cards. A year or two later he bought a graphics card that was twice as fast as before. So for a short moment Peter was happy again and I was happy, until he added more. Then we tried to get a good frame rate – 30 frames per second plus with a lot of compositing. It's a very graphics-heavy tool – to actually have the whole graphically-complex editing and the compositing underneath with multi-outputs to your audience all in one tool is daunting. But Mixxa works. Fortunately I work with our programmers here at Derivative, and Mixxa is a great test case to optimize the speed of TouchDesigner's video playback and compositing.

I'm always designing small prototypes and experimental tools, and then determining, can TouchDesigner do this? For me, I'm lazy – I don't want to program anything, though I don't mind some Python scripting. I want to be able to build everything by just connecting nodes together and adjusting parameters, with a minimum of typing. Within those constraints, our team at Derivative determines what's lacking in TouchDesigner. Then our developers get involved in providing new operators or features, or optimizing the speed of TouchDesigner to reach certain performance goals. It's a lot of back and forth between Peter as a user of a tool and myself and our dev team. He conceives of capabilities he wants, how he wants to organise and perform things, and I am in the middle with my own design ideas – then our programmers help to realise something practical.

SG: It's an interesting concept. I know he is a live visualist, but he's also a filmmaker. So it's interesting to use what is a Live Visual tool, and then to turn it into something that goes into a rendered film.

GH: He had shot lots of video footage for *The End of Time*. And he knew that he was going to mix some of it live, and then record and insert it into the film. He chose about 100 clips that he wanted to mix live. So we converted them into a slightly lower resolution for Mixxa to get a good playback frame rate. He practised a bunch of mixes and ideas at some live events informally, and then it came down to some combinations that he wanted to live mix for the movie. So basically he was still performing live, but this time it was in the studio, working the controls, going from phase to phase, trying to get some timing right while performing along some testbed electronic music he chose for the film. We recorded numerous takes, picked the best ones and rendered them out at full resolution. Those became ProRes clips that ended up in the film. Like I said, this is at the border of what was feasible on a portable gamer's laptop.

SG: That's the same struggle we all have.

GH: I think, for me, I am interested more in real-time performed graphics. The image quality is important, but it's not as important as responsiveness and also getting good framerate, just getting good, natural motion. And if the image quality has to suffer a bit because of it, for me, that's fine. So I've always kind of followed that.

SG: You can't really follow that though, if you're putting it into a theatre.

GH: Agreed! That's why we captured all the performed gestures, then re-rendered it to full resolution clips in non-real-time, then edited it into the film. It was a hybrid performed-post-rendered project.

SG: Great. That's really interesting. Okay, so here's a last question. It's a 'future' question, so you may not be able to answer. What are the future developments you're planning for Derivative, TouchDesigner or any other venture? It might be interesting gigs that you have coming out.

GH: I see only a limited distance in the future because we are building a product with currently available technology and hunches of what people will want in the near future, which in itself is a lot! We're always trying to make TouchDesigner better for more people in the short term. Now it is used in schools to teach graphics, real-time media arts and even math and science. It's used in interactive art projects, media playback systems, show control, visualisation projects, puppeteering in theme parks and more. It's great to see the synergy of all these fields mixing up – this synergy totally drives us, and it is great that one product can serve all these needs, and that people can go off and build their own applications.

In 2005 it was almost unthinkable that a non-programmer could make their own video mixer product, but now you can actually build your own. And that idea of building your own interactive applications with a procedural node-based system, high rendering quality, great video playback and all that. It opens so much up.

Going forward, TouchDesigner has been evolving to interface with a great variety of input devices and protocols to get live data into and out of TouchDesigner. Like Oculus VR and Leap Motion, robotic machinery, Arduinos, DMX lighting control, trackers and all that. That's natural – TouchDesigner is designed to flow and process data through it. More web interaction and more web interoperation as well, and TouchDesigner in the cloud.

I would very much like to be performing more stuff, but as we were discussing earlier, I spend 95% of my time developing the tool and 5% trying it out. I actually like building tools, getting them out to people and letting them take it from there, seeing what they do with it, and informing what our development team builds next. I personally prefer performing in situations where there's zero expectation about what you are about to do, just turn up, plug some gear in, work with the people around you and experiment. So I like the more guerrilla-style situations.

SG: Some people have like a master plan of what they want to do. You guys have obviously come a long way right now, to the point where it's not like you

need to solve that many problems. At some other companies that are really just starting up. They need to solve every problem. So they really need to have a set of milestones for the future.

GH: Yeah. I think we've reached a point where the infrastructure of TouchDesigner is pretty broad and open. It embodies 3D, audio, compositing, user interface building, data interoperability between TouchDesigner and other systems. That's a great starting point. Going forward, we find it most exciting simply keeping pace with the expanding horizons of the TouchDesigner community.

Notes

1 SideFX Official Website, www.sidefx.com/company/about-sidefx/.
2 Norman McLaren, *Synchromy*, National Film Board of Canada Vimeo page, 1971, https://vimeo.com/29399459.
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4 *Omnibus Animation Demo Reel (1985)*, VintageCG youtube page, 2009, www.youtube.com/watch?v=K18ZcE2t1Kw.
5 Ryan Ulyate, David Bianciardi, Judith Crow, and Greg Hermanovic, "Interactive Dance Club '98: A Legend in the Making!," in *ACM SIGGRAPH 2018 Panels* (*SIGGRAPH '18*) (New York, NY, USA: Association for Computing Machinery), Article 3, 1–2. https://doi.org/10.1145/3209621.3214887.
6 Peter Mettler, *The End of Time*, Vimeo on-demand page (paywall), 2018, https://vimeo.com/ondemand/eot.